

- 7(a)** Potential difference used to accelerate the ions, $\Delta V = 1060 - (-225) = 1285 \text{ V}$
The gain in kinetic energy of the ion = its loss of electric potential energy

$$\frac{1}{2}mu^2 = q\Delta V, \text{ where } u \text{ is the speed of the ion. [CONCEPT]}$$

$$u = \sqrt{\frac{2q\Delta V}{m}} = \sqrt{\frac{2(1.60 \times 10^{-19})(1285)}{2.2 \times 10^{-25}}} = 4.323 \times 10^4 \text{ ms}^{-1}$$
$$= 4.3 \times 10^4 \text{ ms}^{-1}$$

- (b)(i)** The passage says it takes 4 days (or more) to eject 1 kg of xenon.
maximum mass of xenon ejected per second,

$$\frac{\Delta m}{\Delta t} = \frac{1 \text{ kg}}{4 \text{ days}}$$

$$\frac{\Delta m}{\Delta t} = \frac{1}{4 \times 24 \times 60 \times 60} = 2.894 \times 10^{-6} \text{ kg s}^{-1} = 2.9 \times 10^{-6} \text{ kg s}^{-1}$$

- (ii) Maximum force exerted on the ejected xenon,

$$\begin{aligned}
 F &= \frac{\Delta p}{\Delta t} \text{ [CONCEPT]} \\
 &= \left(\frac{\Delta m}{\Delta t} \right) u \\
 &= (2.9 \times 10^{-6})(4.3 \times 10^4) \\
 &= 0.1247 \text{ N}
 \end{aligned}$$

By Newton's 3rd law, the maximum thrust = $F = 0.12 \text{ N}$

- (c) Assume that, at the start, the mass of Deep Space 1 (M_{DS}) and the mass of the xenon (m_{Xe}) are at rest. So their total initial momentum $p_{initial} = 0$

Applying the principle of conservation of momentum, [CONCEPT]
their total final momentum $p_{final} = p_{initial} = 0$

So the momentum of DS must be equal and opposite to that of Xe,

$$\begin{aligned}
 M_{DS} v_{DS} &= m_{Xe} v_{Xe} \\
 M_{DS} &= \frac{m_{Xe} v_{Xe}}{v_{DS}} = \frac{(74)(4.3 \times 10^4)}{(4.3 \times 10^3)} = 740 \text{ kg}
 \end{aligned}$$

- (d) The positive ions will be attracted to the negatively charged spacecraft even as the ions are ejected away, so they will be ejected at a lower speed than intended.

- (e) Any 2 reasonable points:

- Ion engines use fuel more efficiently than chemical rockets, so they can travel larger distances per unit mass of fuel. [*Nasa source: Chemical rockets have demonstrated fuel efficiencies up to 35%, but ion thrusters have demonstrated fuel efficiencies over 90%.*]
- Ion engines can produce a larger increase in speed per unit mass of fuel used, compared to chemical rockets, so they can travel longer distances with less fuel.
- Ion engines use their fuel much more slowly, so the fuel can last for much longer periods of time.
- Ion engines use inert gases (like xenon) for propellant and there is no risk of the explosions associated with chemical rockets, so ion engines are less risky (for long missions).

- (f)(i) Energy of a photon, $E = \frac{hc}{\lambda}$ and momentum of a photon, $p = \frac{h}{\lambda}$

where λ is the wavelength of the photon

$$\text{Substitute } p = \frac{h}{\lambda} \text{ into } E = \frac{hc}{\lambda} \text{ gives, } E = \frac{hc}{\lambda} = \left(\frac{h}{\lambda} \right) c = pc$$

['Show' question: need to explain your steps and show substitution of equation for p into equation for E]

- (ii) Power received by 1 m² of sail, power = intensity $\times$ area = 1400 W
Energy received by 1 m² of sail in 1 s, $E = \text{power} \times \text{time} = 1400 \text{ J}$

From $E = pc$, we have

$$\text{Momentum in 1 s, } p = \frac{E}{c} = \frac{1400}{3.00 \times 10^8} = 4.667 \times 10^{-6} \text{ N s} \\ \approx 4.7 \times 10^{-6} \text{ N s}$$

- (iii) total momentum of the photons incident on 32 m^2 of sail in 1 s,

$$p_{\text{total}} = p \times 32 = (4.7 \times 10^{-6})(32) = 1.504 \times 10^{-4} \text{ N s}$$

Change of momentum upon reflection,

$$\Delta p = 2p_{\text{total}} = 2(1.504 \times 10^{-4}) = 3.008 \times 10^{-4} \text{ N s}$$

Force = change of momentum per unit time = $3.008 \times 10^{-4} \text{ N}$

By Newton's 3rd law, the force on the whole sail = $3.0 \times 10^{-4} \text{ N}$

- (iv) If all the light is absorbed, then the final momentum of the light is 0.

So the change of total momentum is half of that in (iii).

Hence, the force on the sail is also halved.

- (g) LightSail orbits Earth with an orbital radius of 7020 km.

The centripetal acceleration of LightSail = the gravitational field strength at that orbit.

[CONCEPT]

$$a_c = \frac{GM}{r^2} = \frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{(7020 \times 10^3)^2} = 8.12 \text{ m s}^{-2}$$

So the centripetal acceleration of LightSail 2 cannot be 0.058 mm s^{-2} .

- (h) Advantage: (or some other reasonable point)

- The solar sail will never run out of "fuel" so long as there is light shining on the sail, whereas ion engines can run out of fuel.

Disadvantage: (any 1 or some other reasonable point)

- The thrust of a solar sail ($3.0 \times 10^{-4} \text{ N}$) is much lower than that of an ion engine (0.12 N).
- The sail is very thin ($4.5 \text{ } \mu\text{m}$) and may be easily damaged by space rocks. (whereas an ion engine is not so fragile)
- The total surface area of a solar sail needs to be very big, but that makes it more susceptible to collisions with space rocks.